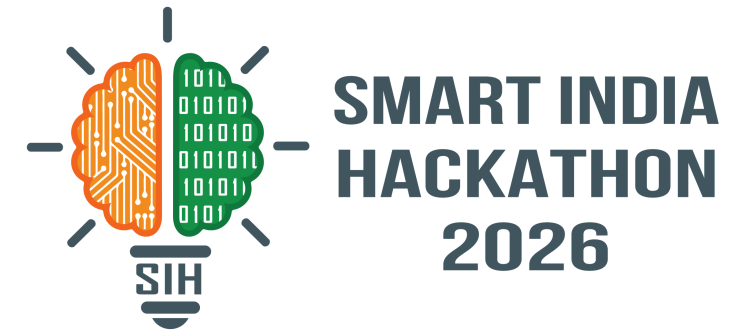
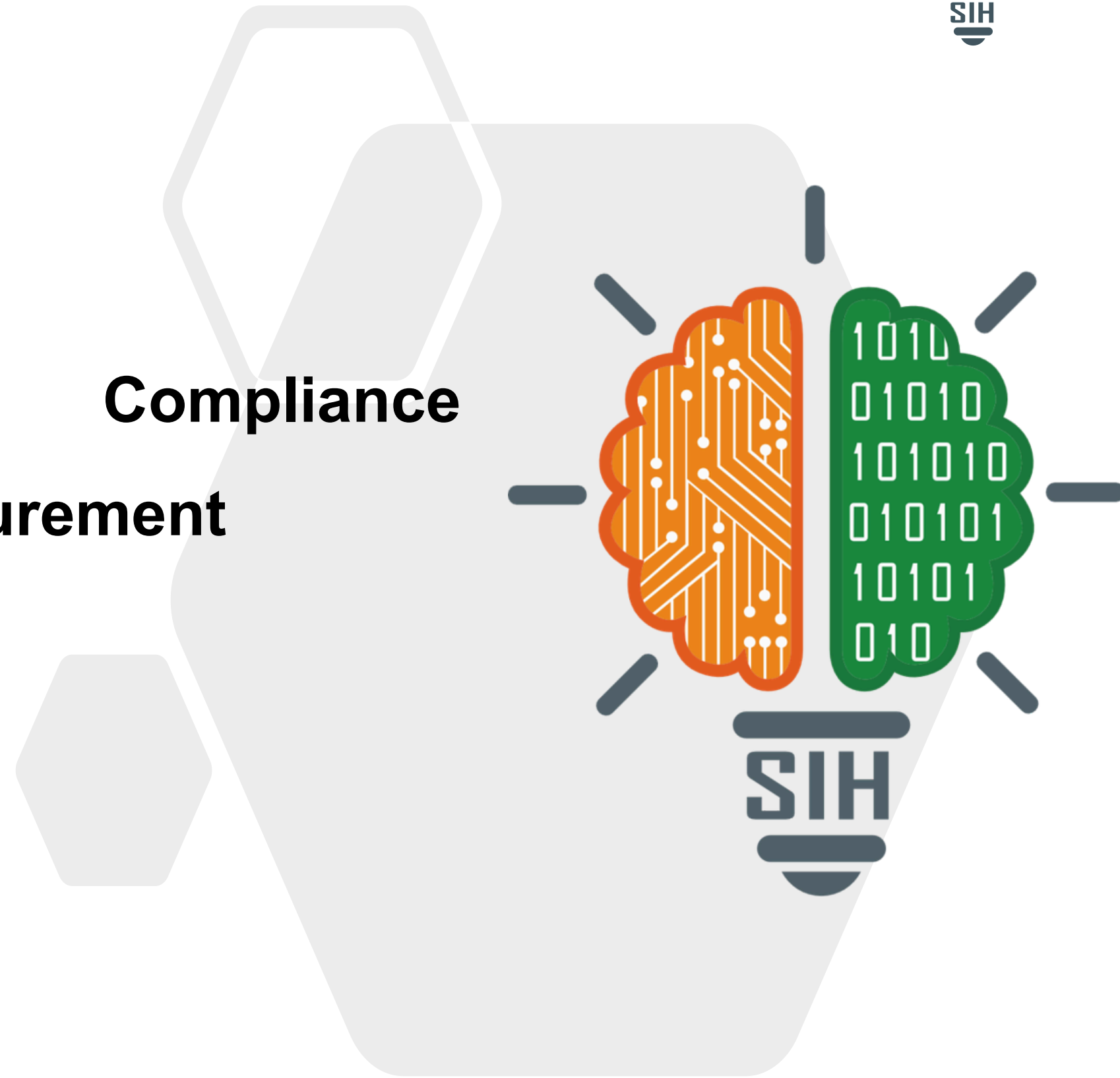


SMART INDIA HACKATHON 2026



- 26100
- AI Powered Integrated Bid Compliance Verification Platform for GeM Procurement
- Software
- Plugged In



IDEA TITLE

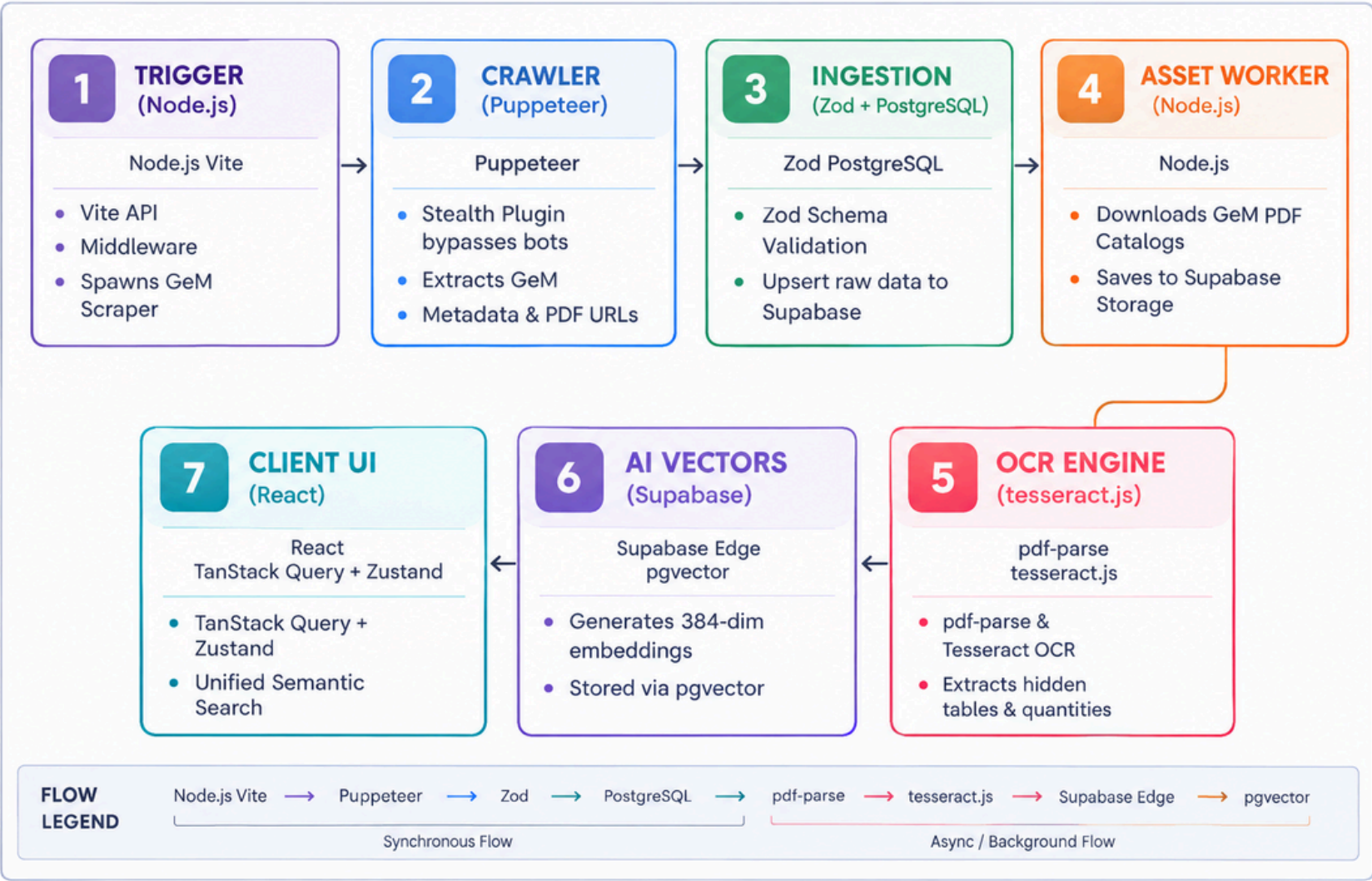
- **Proposed Solution (Describe your Idea/Solution/Prototype)**
- Lelam.co is an AI-powered government auction discovery and assistance platform.
- It finds and analyzes MSTC auctions, extracting catalogue data and applying OCR to scanned documents.
- AI semantic search surfaces opportunities with ROI estimation, watchlists, and logistics management.
- The Compliance Engine evaluates each requirement using configurable, tender-specific rules.
- Verified data, deterministic rules, and supporting evidence create a transparent assessment—not unsupported AI qualification decisions.
- It aggregates GeM, MSTC, and other government and institutional procurement platforms into one marketplace for discovery, participation, search, filtering, and recommendations.

Frontend: React/Angular + TypeScript
Build: Vite 8
UI & Styling: Tailwind CSS
State Management: Zustand
Data Management: TanStack React Query
Backend: Vercel Serverless Functions
Database: Supabase + PostgreSQL
Semantic Search: PostgreSQL pgvector

Authentication & Storage: Supabase
AI Document Processing: Tesseract.js + PDF Parser
Automation & Data Integration: Puppeteer
Data Visualization: Recharts + D3.js

TECHNICAL APPROACH

Ingestion Pipeline



Compliance Engine



1. Feasibility of the Idea

AI-based compliance verification is already technically feasible and similar solutions for document verification and compliance analysis exist.

However, there is a significant gap in having a **single platform that unifies and standardizes procurement and auction opportunities from multiple fragmented platforms.**

The proposed solution combines **unified auction discovery, standardized data, AI-powered compliance verification and bidder eligibility assessment** in one platform.

The modular architecture makes the platform **scalable**, allowing additional auction platforms, government databases and compliance requirements to be integrated over time.

2. Challenges, Risks & How We Overcome Them

Different platforms use different data formats and structures

→ Develop **platform-specific connectors** within a common **data ingestion pipeline** to extract and normalize information into a standardized format.

Auction platforms may change their websites or data structures

→ Build modular ingestion services that can be independently updated without affecting the rest of the platform, while prioritizing official APIs wherever available.

Complex and inconsistent tender documents

→ Use an **AI/OCR document processing pipeline** to extract requirements and convert them into structured, machine-readable compliance rules.

Impact of the Proposed Solution

- **One unified platform** to discover and access opportunities that are currently distributed across multiple platforms.
- Makes government auctions and procurement **easier to discover and more accessible**, helping bring auctions into the mainstream.
- Enables businesses to find opportunities based on their **industry, eligibility, location and verified compliance profile**.
- Reduces the need to repeatedly search through multiple portals and understand different platform interfaces.
- Helps businesses identify **relevant auctions they are eligible to participate in**.
- Reduces repetitive document submission and compliance verification through a reusable verified bidder profile.

Benefits of the proposed solution

- **Unified Procurement Ecosystem:** Aggregates and standardizes opportunities from fragmented auction and procurement platforms into a single interface.
- **Time Saving:** Significantly reduces the time businesses spend searching multiple platforms and manually checking eligibility requirements.
- **Reduced Government Workload:** Automates document analysis, cross-verification and initial compliance screening, reducing the workload on procurement officers.
- **Faster Tender Evaluation:** Helps officers quickly identify compliant, non-compliant and high-risk bids.
- **Reduced Human Error:** Standardized rules and automated cross-verification reduce inconsistencies in manual verification.
- **Improved Transparency:** Provides evidence-backed compliance results and a complete audit trail for verification and decision-making.

ar	Paper	Conference / Journal	Key Contribution
25	A Deep Learning Approach for Identifying Similarities in Contracted Object Description in Public Procurement	Procedia Computer Science (Elsevier)	AI-based procurement document similarity
25	Semantic Search Approach for Classified Platforms	IEEE Signal Processing and Communications Applications Conference (SIU 2025)	Semantic search using vector embeddings for document retrieval.
24	VectorSearch: Enhancing Document Retrieval with Semantic Embeddings and Optimized Search	arXiv (Preprint)	Improved semantic document retrieval using vector embeddings.
24	Nougat: Neural Optical Understanding for Academic Documents	ICLR 2024 (International Conference on Learning Representations)	Transformer-based OCR for structured document understanding